Only one (more) functionality

If you're preparing a hardware device with only one sensor for one functionality, the Arduino™ Nano and the Arduino™ Micro are best suited for such applications. They're also useful if you fall short of pins and need to add just one more functionality to your project by allowing communication between the two microcontrollers, if necessary.

Advice for complete beginners

For fewer sensors and other components, the Arduino™ Uno or Arduino™ Leonardo board are good choices. On the other hand, if you need to connect many sensors and/or communicate with more than one device, we advise you to get the Arduino™ Mega 2560 board, until you get an idea of the right number of pins needed for your application.

Which one do you choose?

Addressing the cost issue

For most of us on a budget, Arduino™ compatible boards, which are clones of original Arduino™ boards, are also available in the market. Do keep in mind that these boards aren't made by Arduino™ makers themselves.

There's another cheaper version of Arduino™ called 'Freeduino'. There are minor differences, which might overwhelm a complete beginner. So a complete beginner should get the clone, if not the original Arduino™.

Finally, if budget is no issue, an original board from Italy is your way to go.

Note: Pins serve multiple functionality

Most pins on an Arduino™ board serve dual functionality and hence it's always advisable to consider that the maximum pins available are Analog In pins, Analog Out pins and Digital I/O pins. You'll notice that the Arduino™ Uno has only 20 pins available directly for use (6 Analog In + 14 Digital I/O pins). Of these 14 digital pins, six also act as PWM pins. So if you need these six pins for PWM output, you only have eight pins for Digital I/O.